

The ultimate

Dataset Bible

for Data Engineers

110+

REAL-WORLD
DATASETS

14

INDUSTRY
CATEGORIES

100%

FREE TO
ACCESS

Stop practicing on toy data. These are the same public sources used to build production pipelines, recommender systems, and streaming platforms at real companies.

Shashank Mishra

Staff Data Engineer · Founder, GrowDataSkills

growdataskills.com

Why this guide exists

Every aspiring data engineer hits the same wall — you understand Spark, Airflow, Kafka, dbt, and Snowflake in theory, but when it's time to build a portfolio project, you end up writing a fake-data generator with `faker` and calling it a day.

Recruiters can smell synthetic data from a mile away. What they actually want to see is how you handle the messiness of the real world — skewed distributions, missing values, late-arriving records, schema drift, and volumes that don't fit in memory.

This guide is a hand-picked collection of public, free, and legal datasets covering every scenario a data engineer needs to practice. Bookmark it. Share it. Build from it.

How to actually use this

- ▶ **Pick one category** that aligns with an industry you want to work in.
- ▶ **Build one project end-to-end** — ingestion → storage → transformation → serving.
- ▶ **Use multiple tools** — cloud storage, a warehouse, an orchestrator, and a BI layer.
- ▶ **Document everything** on GitHub with a proper README and architecture diagram.
- ▶ **Post about it** on LinkedIn — visibility is half the game.

Project ideas by pipeline type

Batch ETL

NYC Taxi · Olist E-commerce · IMDb · World Bank

Streaming

Wikimedia · Binance WS · OpenSky · MTA Realtime

Data Warehouse

TPC-DS · Stack Overflow · GitHub Archive

Analytics / BI

Airbnb · MovieLens · FRED · Online Retail II

API Ingestion

REST Countries · Spotify · GitHub · NewsAPI

Geo / IoT

OpenStreetMap · NOAA · OpenSky · Divvy Bikes

01



CATEGORY 01

General Dataset Repositories

Your first stop — thousands of datasets across every domain

Kaggle Datasets

400K+ community & company-curated datasets covering every domain imaginable. The undisputed king for practice data.

➤ www.kaggle.com/datasets

Google Dataset Search

Google-scale search engine specifically for finding datasets across the open web.

➤ datasetsearch.research.google.com/

UCI Machine Learning Repository

The classic academic repository — clean, benchmark-quality datasets used in thousands of papers.

➤ archive.ics.uci.edu/

Hugging Face Datasets

200K+ datasets with native parquet/arrow loading, perfect for modern ML & data pipelines.

➤ huggingface.co/datasets

Awesome Public Datasets (GitHub)

Community-maintained giant index of free, high-quality public datasets by topic.

➤ github.com/awesomedata/awesome-public-datasets

Papers With Code Datasets

7,000+ ML benchmark datasets, each linked to the papers that use them.

➤ paperswithcode.com/datasets

Data.world

Social data-sharing platform with collaborative datasets, SQL workspaces, and API access.

➤ data.world/

Academic Torrents

Large research datasets (Reddit, MIMIC, etc.) distributed via torrents — huge volumes.

➤ academictorrents.com/

02



CATEGORY 02

Cloud Provider Public Datasets

Huge, production-grade data already sitting in S3, GCS & Azure Blob

AWS Open Data Registry

Petabyte-scale datasets hosted on S3 — NOAA, NASA, Common Crawl, OpenStreetMap & more.

➤ registry.opendata.aws/

Google Cloud Public Datasets

Huge BigQuery-ready datasets — GitHub, Wikipedia, Stack Overflow, Chicago Taxi, Ethereum.

➤ cloud.google.com/datasets

BigQuery Public Datasets

Query terabytes of data for free directly from BigQuery — goldmine for SQL practice.

➤ console.cloud.google.com/marketplace/browse?filter=solution...

Azure Open Datasets

Curated datasets integrated with Azure ML & Synapse — NYC Taxi, weather, OSM, demographics.

➤ learn.microsoft.com/en-us/azure/open-datasets/dataset-catalog

Databricks Datasets

Sample datasets mounted at /databricks-datasets/ — perfect for Spark/Delta practice.

➤ docs.databricks.com/en/dbfs/databricks-datasets.html

Snowflake Marketplace

Hundreds of live, query-ready datasets you can share into your Snowflake account.

➤ app.snowflake.com/marketplace

03



CATEGORY 03

Government & Open Data

Authentic, large-scale data straight from official sources

Data.gov (USA)

300,000+ datasets from every US federal agency — census, energy, agriculture, transportation.

➤ www.data.gov/

Data.gov.in (India)

India's official open-data portal — 600K+ resources across ministries and states.

➤ data.gov.in/

Data.gov.uk (UK)

UK government's open data — spending, crime, transport, health.

➤ data.gov.uk/

EU Open Data Portal

Datasets from 90+ EU institutions and agencies in every domain.

➤ data.europa.eu/

World Bank Open Data

Global development indicators for every country going back 60+ years.

➤ data.worldbank.org/

UN Data

Official stats from UN agencies — demographics, labour, trade, environment.

➤ data.un.org/

WHO Global Health Observatory

Country-level health data on diseases, mortality, immunization, and more.

➤ www.who.int/data/gho

IMF Data

Macroeconomic & financial data — GDP, inflation, balance of payments across 190+ countries.

➤ www.imf.org/en/Data

OECD Data

Economic, social, and environmental indicators for OECD member countries.

➤ data.oecd.org/

NYC Open Data

One of the richest city datasets in the world — 311, crime, permits, taxi, schools.

➤ opendata.cityofnewyork.us/

Chicago Data Portal

Famous for the Chicago Crime & Taxi datasets used in countless tutorials.

➤ data.cityofchicago.org/

MOSPI India

India's Ministry of Statistics — NSS surveys, CPI, IIP, and national accounts.

➤ mospi.gov.in/

04



CATEGORY 04

Finance & Economics

Build trading dashboards, forecasting pipelines & market analytics

Yahoo Finance (via yfinance)

Free Python library for historical stock, ETF, crypto, and index price data — no API key needed.

➤ pypi.org/project/yfinance/

Alpha Vantage API

Free tier gives real-time & historical stock, forex, crypto data with technical indicators.

➤ www.alphavantage.co/

Nasdaq Data Link (Quandl)

Curated financial & economic datasets from hundreds of publishers — many are free.

➤ data.nasdaq.com/

FRED (Federal Reserve)

800,000+ US & global economic time series — GDP, unemployment, interest rates.

➤ fred.stlouisfed.org/

SEC EDGAR

Every 10-K, 10-Q, and filing from every US public company — gold for NLP & finance projects.

➤ www.sec.gov/edgar.shtml

RBI Database (India)

Reserve Bank of India's DBIE — banking, forex, money market, and payment system data.

➤ dbie.rbi.org.in/

NSE / BSE Historical Data

Indian stock market historical data downloadable from the exchange websites.

➤ www.nseindia.com/reports-indices-historical-index-data

CoinGecko API

Free crypto API with prices, market caps, volume for 10,000+ coins & 1,000+ exchanges.

➤ www.coingecko.com/en/api

Binance Public API

Live crypto tickers, klines, order books — perfect for real-time streaming projects.

➤ binance-docs.github.io/apidocs/

World Bank Financial Inclusion

Global Findex — how adults save, borrow, and make payments across 140+ countries.

➤ www.worldbank.org/en/publication/globalfindex



E-commerce & Retail

Recommendation engines, customer analytics & RFM segmentation

Amazon Reviews Dataset

233M Amazon reviews across 29 product categories — huge for NLP & recommender systems.

➤ nijianmo.github.io/amazon/index.html

Instacart Market Basket

3M grocery orders from 200K users — the classic market basket analysis dataset.

➤ www.kaggle.com/c/instacart-market-basket-analysis/data

Olist Brazilian E-commerce

100K orders with multi-table relational schema — perfect for modeling a star schema.

➤ www.kaggle.com/datasets/olistbr/brazilian-ecommerce

Online Retail II (UCI)

1M+ transactions from a UK retailer — the go-to dataset for RFM & cohort analysis.

➤ archive.ics.uci.edu/dataset/502/online+retail+ii

H&M Personalized Fashion

Transaction, article & customer data from H&M — 31M rows. Great for recommender pipelines.

➤ [www.kaggle.com/competitions/h-and-m-personalized-fashion-re...](https://www.kaggle.com/competitions/h-and-m-personalized-fashion-recommendation)

Retailrocket Recommender

Real visitor behavior — views, add-to-carts, transactions from an e-commerce site.

➤ www.kaggle.com/datasets/retailrocket/ecommerce-dataset

Flipkart Products Dataset

20K+ product listings from Flipkart — useful for catalog, NLP & pricing analytics.

➤ www.kaggle.com/datasets/PromptCloudHQ/flipkart-products

06



CATEGORY 06

Transportation & Mobility

The absolute gold standard for data pipeline & streaming projects

NYC Taxi & Limousine Trip Data

THE classic DE dataset — monthly Parquet files of every yellow/green cab trip since 2009.

➤ www.nyc.gov/site/tlc/about/tlc-trip-record-data.page

Uber Movement

Anonymized travel-time data across cities — great for geospatial analytics.

➤ movement.uber.com/

US DOT Flight Delays (BTS)

Every US domestic flight with delays, cancellations, carriers — decades of history.

➤ www.transtats.bts.gov/

OpenFlights Database

Airports, airlines, and routes worldwide — CSV downloads, small but high quality.

➤ openflights.org/data.html

GTFS Transit Feeds

Standardized public-transit schedules from thousands of agencies worldwide.

➤ gtfs.org/resources/data/

Inside Airbnb

Airbnb listings, reviews, and calendar data scraped for cities across the world.

➤ insideairbnb.com/get-the-data/

OpenStreetMap Data

Full planet dump or regional extracts — maps, POIs, roads. Massive and free.

➤ download.geofabrik.de/

Divvy Bikes (Chicago)

Every single Divvy bike-share trip since 2013 — perfect for time-series & geospatial.

➤ divvybikes.com/system-data

Citi Bike (NYC)

Monthly trip history for NYC's bike-share program.

➤ citibikenyc.com/system-data

07



CATEGORY 07

Healthcare & Life Sciences

Real clinical & public-health data for impactful projects

MIMIC-IV (PhysioNet)

De-identified ICU records for 300K+ patients. Free after a short credentialing course.

➤ physionet.org/content/mimiciv/

CDC WONDER

US public-health queryable database — mortality, natality, environmental.

➤ wonder.cdc.gov/

HealthData.gov

3,000+ US health datasets from CMS, FDA, CDC.

➤ healthdata.gov/

Our World in Data

Beautifully curated global data on health, development & environment — CSV downloads.

➤ ourworldindata.org/

JHU COVID-19 Data

The famous daily COVID case/death time-series that powered the global dashboards.

➤ github.com/CSSEGISandData/COVID-19

NHS Digital (UK)

Official UK health data — hospital stats, prescriptions, outcomes.

➤ digital.nhs.uk/data-and-information

Kaggle Healthcare Datasets

Curated tag page with hundreds of clinical & medical datasets.

➤ www.kaggle.com/datasets?tags=13201-Healthcare



Web, Social & Text Data

Massive unstructured data for NLP, LLM & log-processing projects

GitHub Archive (GHArchive)

Every public GitHub event since 2011 — commits, PRs, issues. Query via BigQuery or download JSON.

➤ www.gharchive.org/

Common Crawl

Petabytes of crawled web pages. The data that trained GPT, LLaMA & every major LLM.

➤ commoncrawl.org/

Wikipedia Dumps

Full Wikipedia database dumps in every language — refreshed monthly.

➤ dumps.wikimedia.org/

Wikimedia Pageviews

Hourly pageview counts for every Wikipedia article — amazing for trend analysis.

➤ dumps.wikimedia.org/other/pageviews/

Stack Exchange Data Dump

Every question, answer & vote on Stack Overflow + all Stack Exchange sites. XML dumps.

➤ archive.org/details/stackexchange

Reddit Submissions & Comments

Historical Reddit archives available via Academic Torrents.

➤ academictorrents.com/browse.php?search=reddit

GDELT Project

A global event database — monitors news in 100+ languages every 15 minutes.

➤ www.gdeltproject.org/

HackerNews BigQuery Dataset

Every HackerNews story & comment ever. Already loaded in BigQuery public datasets.

➤ console.cloud.google.com/marketplace/product/y-combinator/h...



Movies, Music & Books

Build recommender systems, trend dashboards & graph analytics

IMDb Non-Commercial Datasets

Official TSV dumps of titles, ratings, cast, crew — refreshed daily.

➤ developer.imdb.com/non-commercial-datasets/

MovieLens Datasets

25M movie ratings from GroupLens — the canonical recommender-system dataset.

➤ grouplens.org/datasets/movielens/

TMDb API

Free movie & TV metadata API — posters, cast, ratings, trailers.

➤ developer.themoviedb.org/

OMDb API

Simple free API for movie info pulled from IMDb.

➤ www.omdbapi.com/

Spotify Web API

Track, artist, album, and audio-feature data. Great for music analytics.

➤ developer.spotify.com/documentation/web-api

Million Song Dataset

Audio features & metadata for 1M songs — classic music-ML dataset.

➤ millionsongdataset.com/

Last.fm Dataset

Listening histories linked to the Million Song Dataset.

➤ millionsongdataset.com/lastfm/

Open Library Dumps

Full catalog of Open Library — books, authors, editions as JSONL.

➤ openlibrary.org/developers/dumps

Goodreads Books Dataset (Kaggle)

Popular 10K-book dataset with ratings — simple entry point for book analytics.

➤ www.kaggle.com/datasets/jealousleopard/goodreadsbooks



Sports Analytics

Match events, ball-tracking & historical stats across every sport

StatsBomb Open Data

Free event data for football matches — Messi's La Liga career, World Cups, Euros.

➤ github.com/statsbomb/open-data

Cricsheet

Ball-by-ball data for every international & IPL cricket match. JSON & YAML.

➤ cricsheet.org/

football-data.co.uk

Decades of football match results & betting odds across all major leagues. Free CSVs.

➤ www.football-data.co.uk/

Retrosheet (Baseball)

Play-by-play MLB data back to 1901 — incredibly detailed.

➤ www.retrosheet.org/

NBA Stats API (unofficial)

Free Python wrapper around the nba.com endpoints — every shot, play, box score.

➤ github.com/swar/nba_api

FIFA Players Dataset (Kaggle)

Every FIFA game's full player ratings — fun for ranking & clustering projects.

➤ www.kaggle.com/datasets/stefanoleone992/fifa-22-complete-pl...

F1 Ergast API

Full Formula 1 historical data — races, drivers, lap times, pit stops.

➤ ergast.com/mrd/



Weather & Climate

Time-series heaven — forecasting, anomaly detection & IoT simulations

NOAA Climate Data Online

Massive archive of US & global weather observations — daily, hourly, by station.

➤ www.ncei.noaa.gov/cdo-web/

Open-Meteo API

Completely free, no-key weather API with historical + forecast data worldwide.

➤ open-meteo.com/

OpenWeatherMap API

Popular free-tier weather API — current, forecast, and historical.

➤ openweathermap.org/api

NASA Earthdata

Earth-observation data — satellite imagery, MODIS, Landsat. Enormous.

➤ earthdata.nasa.gov/

Copernicus Climate Data Store

ERA5 reanalysis — the gold standard of historical climate datasets.

➤ cds.climate.copernicus.eu/

IMD (India Met Dept)

Official weather data for India — rainfall, temperature, cyclones.

➤ mausam.imd.gov.in/

Kaggle Historical Weather

Pre-cleaned city-level historical weather datasets ready to plug in.

➤ www.kaggle.com/datasets?search=weather

12



CATEGORY 12

Real-Time & Streaming Sources

For Kafka, Kinesis, Flink & Spark Streaming projects

Wikimedia EventStreams

Live stream of every edit happening on Wikipedia right now. Free, no auth, SSE.

➤ stream.wikimedia.org/v2/stream/recentchange

Binance WebSocket Streams

Live crypto trades, klines, depth — massively high throughput, perfect for Kafka demos.

➤ [binance-docs.github.io/apidocs/spot/en/#websocket-market-st...](https://binance-docs.github.io/apidocs/spot/en/#websocket-market-streams)

Coinbase WebSocket Feed

Real-time L2 order book and trade data. Great for event-driven pipelines.

➤ docs.cdp.coinbase.com/exchange/docs/websocket-overview

Reddit Live API

Real-time submissions/comments via PRAW or Pushshift.

➤ www.reddit.com/dev/api/

OpenSky Network API

Live and historical ADS-B flight tracking data — plane positions worldwide.

➤ opensky-network.org/apidoc/

MTA / Transit Real-Time Feeds

NYC subway live train positions via GTFS-Realtime. Pure streaming gold.

➤ api.mta.info/

USGS Earthquake Feed

Real-time earthquake events updated every minute in GeoJSON.

➤ earthquake.usgs.gov/earthquakes/feed/



Public APIs for Pipeline Practice

Perfect for end-to-end ETL/ELT pipelines with Airflow, dbt & Snowflake

REST Countries

No-auth API with data on every country — great first API ingestion project.

➤ restcountries.com/

JSONPlaceholder

Fake REST API for prototyping. Posts, users, comments, todos.

➤ jsonplaceholder.typicode.com/

PokeAPI

500MB+ of structured Pokemon data via REST — surprisingly rich schema.

➤ pokeapi.co/

TheMealDB / TheCocktailDB

Free recipe APIs — great for joining & enrichment pipelines.

➤ www.themealdb.com/api.php

NewsAPI

Aggregate headlines from 150K+ sources worldwide (free tier available).

➤ newsapi.org/

GitHub REST API

Pull commits, issues, repos — great for activity analytics dashboards.

➤ docs.github.com/en/rest

Rick and Morty API

Fun, free GraphQL/REST API with characters, episodes & locations.

➤ rickandmortyapi.com/

Public-APIs List

Giant curated list of 1,400+ free public APIs across every domain.

➤ github.com/public-apis/public-apis



Big Data & Warehouse Benchmarks

Scale-test Spark, Snowflake, BigQuery & Databricks clusters

TPC-H

Industry-standard benchmark — 22 complex analytical SQL queries. Scalable from 1GB to 100TB.

➤ www.tpc.org/tpch/

TPC-DS

The modern benchmark — 99 queries simulating a retail data warehouse. Ideal for Spark/Snowflake.

➤ www.tpc.org/tpcds/

Criteo 1TB Click Logs

24 days of actual ad-click data — over 4 billion rows. THE dataset for big-data practice.

➤ ailab.criteo.com/download-criteo-1tb-click-logs-dataset/

NYC Taxi (10+ years)

300M+ rows of taxi trips — perfect for partition/cluster testing on any warehouse.

➤ www.nyc.gov/site/tlc/about/tlc-trip-record-data.page

Stack Overflow BigQuery Dataset

Every post since 2008, pre-loaded in BigQuery — costs pennies to query.

➤ console.cloud.google.com/marketplace/product/stack-exchange...

YOU MADE IT TO THE END

Now go build something **real.**

Knowing 100 datasets is useless if you build from zero of them. Pick one today. Spin up a pipeline. Ship the repo.

Want structured, industry-grade projects to walk you through it?

GrowDataSkills programs are built around real-world projects, modern stacks, and mentorship — not theory videos. Join thousands of learners who transitioned into Data & AI roles.

→ growdataskills.com

CONNECT WITH SHASHANK

- ▶ YouTube youtube.com/@shashank_mishra
- ▶ LinkedIn linkedin.com/in/shashank219
- ▶ Instagram [@_shashank_219](https://instagram.com/_shashank_219)

Found this useful?

Share this PDF with one aspiring data engineer today. Careers are built on lists like this one.